Two cheap rho jump tables compared: chained (main) against the ladder table (v5, v6)

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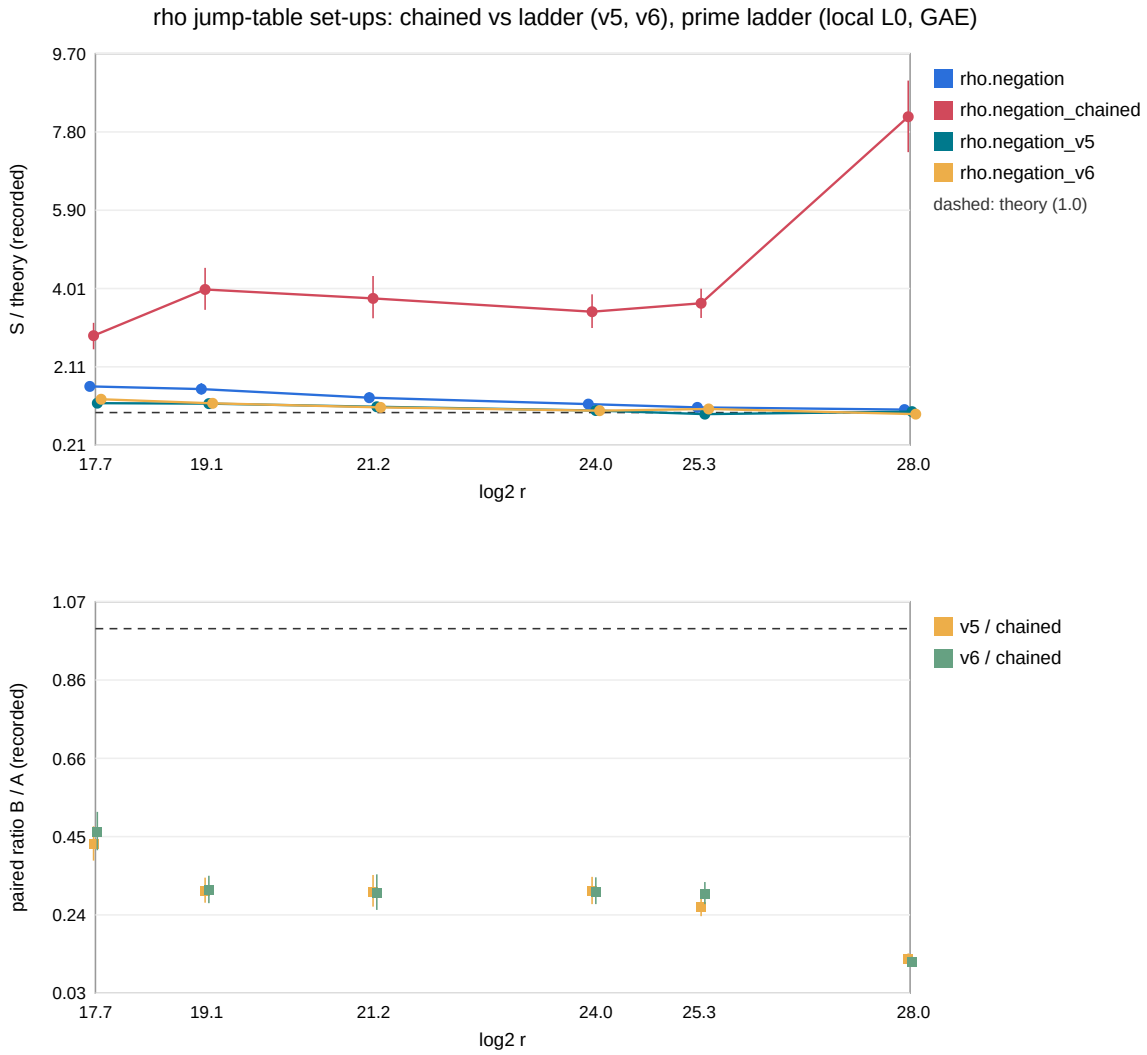
Result (measured, counted GAE, local L0 host) [pcc:unverified]. `rho.negation_chained` (from `research/curve25519_engine_items_20261110`) and this lab's `rho.negation_v5` / `rho.negation_v6` all build the negation-map rho jump table more cheaply than `rho.negation`. They were paired here on six prime curves, $2^{17.7–2}28$, with 32 targets × 6 rounds (1152 pairs per comparison) and an A/A control.

comparison	paired ratio [95 %]	band (w = 0 / 0.1 / 1)
v5 / chained	**0.256 [0.236, 0.277]**	0.256 / 0.257 / 0.261
v6 / chained	**0.261 [0.240, 0.284]**	0.261 / 0.263 / 0.269
chained / rho.negation	3.25 [2.98, 3.54]	3.25 / 3.28 / 3.42
v5 / rho.negation	0.831 [0.797, 0.865]	0.831 / 0.841 / 0.894
v6 / rho.negation	0.849 [0.814, 0.885]	0.849 / 0.861 / 0.919
chained A/A	1.0000	—

Every order holds across the §5a band. The bounds (fit version 2, 6 sizes):

bound	id	ops × floor	α
rho.negation_v5	`ECBND1hbd448fdd4fba`	1.103	0.461
rho.negation_v6	`ECBND1h61c3456f18f0`	1.127	0.451
rho.negation	`ECBND1h434d5b9180f8`	1.328	0.426
rho.negation_chained	`ECBND1ha265c33c87d3`	4.312	0.597



S / theory per size (bound records) and the paired ratios against chained

Why: the same set-up saving, opposite walks

The counters below are per √r, as recorded means:

arm	set-up (adds + doublings)	walk operations	cycle escapes	look-ahead additions
rho.negation	0.122	1.009	0.003	0.051
rho.negation_chained	0.051	**3.708**	**0.045**	0.167
rho.negation_v5	0.046	0.884	0.000	0.011

Both cheap tables cut the set-up by the same amount, to about 0.05 √r. The chained table's jumps satisfy `M_j = M_{j−1} + M_{j−2}`, a three-term integer relation. Under the negation map that becomes a short fruitless cycle: cycles and escapes rise 15× and the walk runs 3.7× longer. The v5 table refuses every three-term relation among its jumps before spending an operation (`c = x ± y`, `c = y − x`, `c = 2x`, `2c = x`), and it uses k ≈ √F jumps instead of 4–16. Its walk is at 1.00 the floor and records no cycle.

This agrees in direction with the chained method's own study, which found 1.532× rho.negation at 20 bits and 2.738× at 24 bits and rejected it. Here the ratio is 3.25× pooled, and 8.2× theory at 2^28.

Run

item	value
spec / session	`spec.json` / `ECBS1h56930a35f003`, 6720 records, all verified

item	value
commit / binary	`c7c4c2874f` (main, clean tree) / sha256 `55289451c555…`
audit	`audit.json`: replay-all 5760 reproduced, 0 problems
isolation	L0 container; counted GAE only